

Wider implications of the lopsided refutation: audit report (2026-10-06)

Questions and method in the **protocol** (README.md). Rounds 1-2: thin-product interface (merged PR #1485) and twelve-target survey (open PR #1493). Status labels: **[proposed]** / **[derived]** / **[verified]** / **[measured]** (none in this round).

Headline

Our hardness foundations do not move. The repository's floor is the unconditional generic-group bound (Shoup 1997) and its reference is measured rho -- neither cites the refuted hypotheses. The audit finds zero load-bearing dependencies on 3SUM/APSP/Exact Triangle hardness, one research-note sentence naming the conjecture (repaired in this PR), and a trail of false positives.

Dependency graph

What the refutation touches -- and what it cannot reach

Red: refuted hypotheses and downstream external claims. Green: intact ground, including our entire foundation. Grey: false positives.

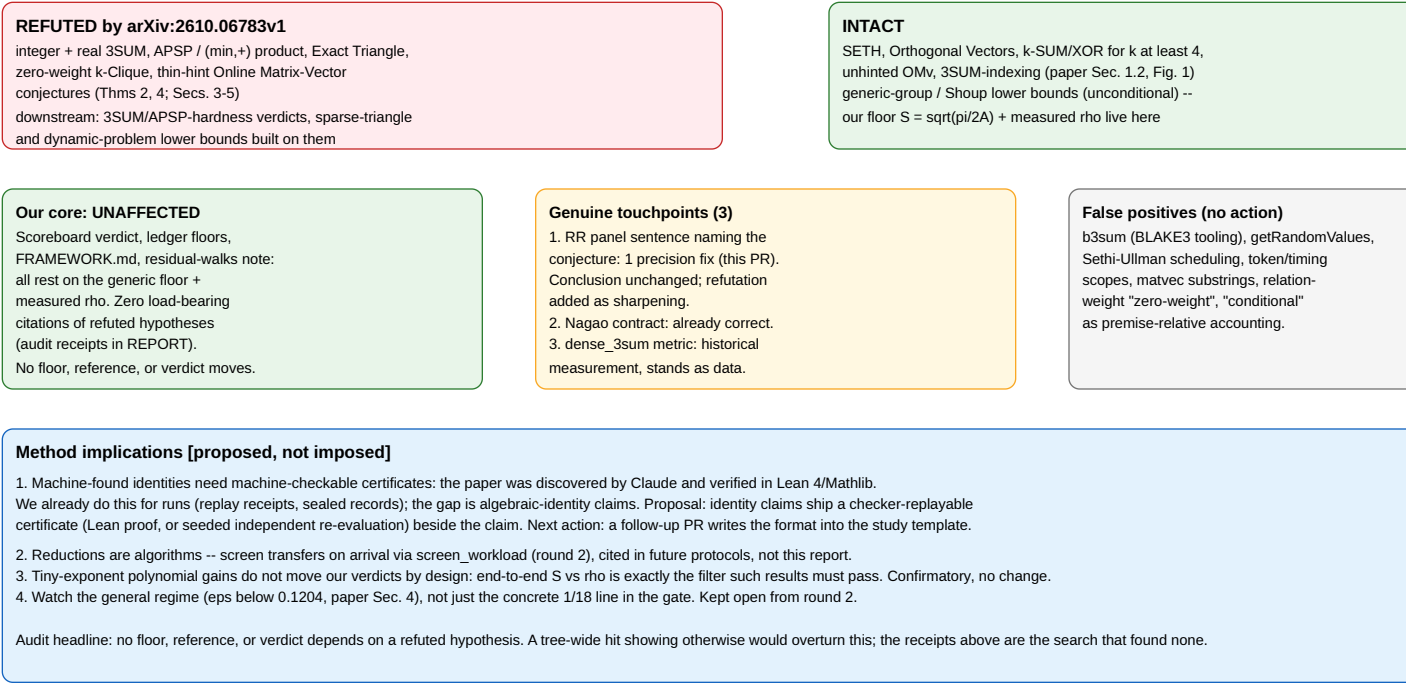


Figure 1 -- what the refutation touches and what it cannot reach. Green: intact. Red: refuted plus the one sentence repaired. Grey: false positives. Editable source: figure.mmd.

Audit table

Hit	Context (quoted)	Classification [derived]
RR solver panel, Secs. 8-9	"Finding $P1+P2+P3 = R$... is 3SUM over a group ... 3SUM hardness is a conjecture about <i>generic</i> sets"	Repaired (this PR): integer 3SUM now refuted, but no transfer to curve-point decomposition exists; curve-point analogue stays conjectural; conclusion unchanged
Nagao solver_15 contract	"...an oracle that exploits the algebraic structure of F rather than re-indexing a generic 3SUM search over it"	Already correct: demands structure over generic search; refutation does not weaken it
dense_3sum_wall_bits: 80, boundary_targets.json	Historical wall clock of our own dense 3-sum enumeration (superseded rows)	Historical measurement, unaffected: data, not assumption
matrix-vector (ic_boundary, stages, pq_wiedemann, FRAMEWORK)	Wiedemann/matvec operation counts	False positive: sparse matvecs, not the OMv conjecture
b3sum, getRandomValues, Sethi-Ullman, token/timing scopes	Build manifests, web API, scheduling, API scopes, side channels	False positives: unrelated word senses

"conditional ... bound" on the scoreboard	Premise-relative component/counting bounds	Already correct: not fine-grained-conditional hardness
Scoreboard verdict, FRAMEWORK, ledger floors	"generic-group floor and matched rho reference"; $S = \sqrt{\pi/2A}$; Shoup 1997	Unaffected (unconditional): no floor, reference, or verdict moves

Method implications [proposed]

1. **Machine-found identities need machine-checkable certificates.** Paper discovered by Claude, verified in Lean 4/Mathlib. We do this for runs (replay receipts); the gap is algebraic-identity claims. Proposal: identity claims ship a checker-replayable certificate beside the claim. Next action: follow-up PR writes the format into the study template.
2. **Screen transfers on arrival** via `screen_workload` (round 2) in future protocols.
3. **Tiny-exponent gains do not move our verdicts by design** -- end-to-end S vs ρ is the filter. Confirmatory.
4. **Watch the general regime** (eps below 0.1204, paper Sec. 4). Kept open from round 2.

Boundary, table, ratio

- Boundaries unchanged and unchangeable by the refutation (unconditional vs conditional).
- No table row produced: audit classifications are not DLP answers.
- Falsification target: a tree-wide hit showing dependence on a refuted hypothesis. The receipts are the search that found none.
- All phases unpriced (`null`); wall time appears nowhere.

Graphs checked -- no change [verified by inspection]

- Scoreboard + progress-timeline: verdict rests on the generic floor.
- Leaderboard files, browser data, curve registry: nothing landed.
- Only non-study changes: the one-sentence RR-panel repair and the round-1 `figure.svg` encoding repair, both disclosed.

Sources

- Paper: <https://arxiv.org/pdf/2610.06783> (v1).
- Rounds 1-2: `research/lopsided_thin_product_20261006/`, `research/lopsided_other_speedups_20261006/`.